SAIPRAGATHEESWAR G

Data Scientist

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LinkedIn Profile

Summary

Results-driven Machine Learning Engineer with a Master's in Data Science and 1.5+ years of experience in developing end-to-end AI/ML solutions. Expertise in fine-tuning large language models (LLMs), building recommendation systems, AI agents, and deploying NLP applications. Skilled in optimizing model performance using PEFT, LoRA, and Langchain. Experienced in creating scalable pipelines, interactive dashboards, and AI-powered tools. Proven ability to deliver impactful, production-ready solutions across diverse domains using cutting-edge technologies.

Experience

•Junior Machine Learning Engineer *September 2024 - Present*

Jobiak Vishakapatnam

- **Salary Prediction using Fine-Tuned LLaMA-based Models with Reinforcement Learning and Human Feedback**
  - Fine-tuned a LLaMA3.1 model with 4-bit quantization to predict salary estimates based on job titles and locations, improving accuracy and memory efficiency.
  - Initially used PEFT with LoRA for model optimization; switched to Reinforcement Learning with Human Feedback (RLHF) after fine-tuning did not meet performance expectations, enabling dynamic adaptation to the task.
  - Employed 4-bit quantization for reduced memory usage, allowing training on large models with limited hardware. The model was trained using the `SFTTrainer` class with gradient accumulation and mixed precision training.
  - **Technologies & Libraries:** Hugging Face Transformers, unsloth, PEFT, LoRA, RLHF, pandas, LLAMA.
- **Job Title Recommendation Using Agents**
  - Built a job title recommendation system using Langchain, Qdrant, OpenAI, and SentenceTransformer to recommend relevant job titles based on user input and search volume.
  - Implemented a Streamlit app for user interaction, providing job title suggestions and high-volume keyword recommendations.
  - Leveraged Langgraph for workflow orchestration and Langsmith for traceability, ensuring smooth function execution and detailed tracking.
  - Utilized dynamic keyword filtering to avoid redundant job titles, ensuring unique and contextually relevant results.
  - **Technologies:** Python, Langchain, Langgraph, Langsmith, Qdrant, OpenAI, SentenceTransformer, Streamlit.

•AI Software Developer *July 2023 - September 2024*

SproutsAI Online

- **Slack App Chatbot Development**
  - Developed a Slack chatbot for daily updates and task management, using Langchain to generate tailored responses from a database.
  - **Tools:** Langchain, MongoDB, Python, Flask.
- **Adaptive Question Selection**
  - Implemented AI for question selection, reducing time by 80%. Collected candidate data, converted profiles to embeddings with a custom OpenAI model, and used ClickHouse for unique question ranking based on candidate data. Deployed as an API with FastAPI and managed using Docker.
  - **Tools:** FastAPI, ClickHouse, OpenAI, Docker, GitHub.
- **NLP Model Deployment**
  - Standardized an NER model with spaCy for job description parsing, enhancing skill extraction.
  - Developed a pipeline to improve job posting visibility on Google for Jobs.
  - **Tools:** NER model, Flask API, BERT, spaCy.
- **Data Scraping**
  - Developed a Chrome extension for candidate data scraping, saving 90% of manual collection time. Designed a user-friendly interface for easy task initiation.
  - Mitigated LinkedIn's anti-scraping measures with adaptive techniques to ensure uninterrupted data collection.
  - **Tools:** JavaScript, HTML/CSS, Chrome Extension, GitHub.
- **Interactive Dashboard Development**
  - Created an interactive dashboard that improved data-driven decision-making efficiency by 25% with real-time visualization features. Used Grafana for visualization and MongoDB for data management.
  - **Tools:** Grafana, MongoDB, JavaScript, HTML/CSS.

Education

- **Master of Technology in Data Science** *2022-24*
  Defence Institute of Advanced Technology, Pune CGPA: 8.38

- **Bachelor of Engineering in Mechanical Engineering** *2017-21*
  University College of Engineering Nagercoil CGPA: 8.23

Personal Projects

- **Smart PDF Chatbot: Your Interactive PDF Companion** *August 2024*
  Built a chatbot system capable of answering user queries by processing custom documents
  - Developed a system to upload PDF documents, split them into chunks, and store them in a vector database using embedding techniques for precise information retrieval.
  - Integrated LangChain, Streamlit, and Hugging Face APIs for embedding models, allowing for intelligent retrieval of relevant chunks based on user queries.
  - Used Groq API and Ollama for open-source LLMs to process and refine the retrieved chunks, delivering accurate and context-aware responses.
  - Implemented memory management to retain conversation history and maintain context over multiple interactions.
  - Deployed the system on Streamlit, creating an interactive and user-friendly interface for seamless question-answer interaction.
  - Technology Used: Python, LangChain, Streamlit, Groq API, Hugging Face, Ollama, Vector Databases

- **Context-Aware Next Word Prediction Using RNN, LSTM, and GRU** *July 2024*
  Developed a next-word prediction model using RNN and deployed it with Streamlit for real-time predictions.
  - Implemented a next-word prediction system utilizing RNN, LSTM, and GRU models to generate contextually relevant word suggestions.
  - Compared and evaluated the performance of each model in terms of accuracy, speed, and contextual understanding.
  - Deployed the model using Streamlit, allowing users to input text and receive real-time next-word predictions based on the models.
  - Technology Used: Python, TensorFlow, RNN, LSTM, GRU, Streamlit

- **Object Detection and Detail Extraction Bot for Visually Impaired Individuals** *April 2024*
  A real-time object detection and detail extraction bot to help visually challenged bots do their daily routine.
  - Implemented 3D object dimension detection in real-world environments, achieving an accuracy rate of 83% for object scale assessment.
  - Created a distance prediction model using obtained 3D data, achieving a precision of 92% in estimating object-to-camera distances.
  - Utilized motion tracking techniques based on object center point localization to analyze and monitor object movements in video frames.
  - Trained a transformer model to predict the trajectory and direction of object movement, enhancing spatial understanding and predictive capabilities.
  - Technology Used: Python, YOLOV8, Tensorflow, PyTorch, Transformers

Technical Skills and Interests

Languages: Python, C, HTML, CSS, Javascript, SQL
Libraries: OpenCV, Numpy, Pandas, Tensorflow, PyTorch, Matplotlib, Playwright, NLTK
Web Dev Tools: Django, VScode, Git, Github, FastAPI, Postman API, Streamlit
Data Visualization: Tableau, Power BI
Cloud/Databases: AWS S3, MySQL, Clickhouse, Milvus, MongoDB, PostgreSQL
Data Engineering: Hadoop framework, Spark
AI Frameworks: LangChain, Hugging Face
Areas of Interest: Data analysis, Data scientist, ML engineer, GenAI engineer